

EE227 – Digital Logic Design

Lecture 18

Outline

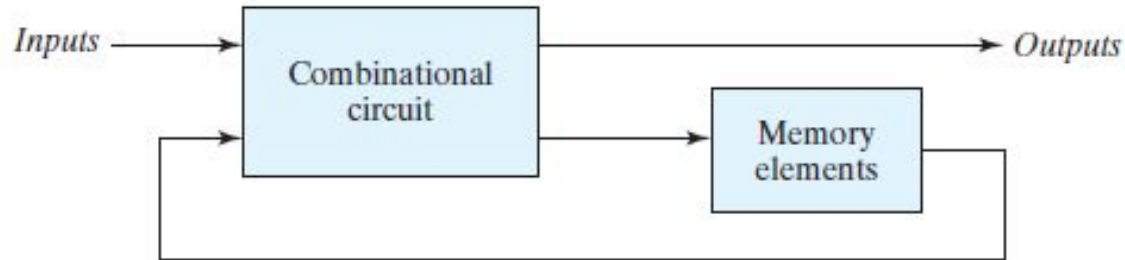
- Sequential Circuits
- Types of Sequential Circuits
 - Synchronous Sequential Circuits
 - Asynchronous Sequential Circuits
- Storage Elements
 - Latches
 - Flip Flops
- SR Latch
- D Latch

Sequential Circuits (Background)

- Today's digital electronic devices have the ability to send, receive, store, retrieve, and process information represented in a binary format
- The technology enabling and supporting these devices is critically dependent on electronic components that can store information
- The digital circuits considered thus far have been combinational and don't have any memory
- Sequential circuits, however, act as storage elements and have memory
- They can store, retain, and then retrieve information when needed at a later time

Sequential Circuits (1/2)

- Sequential circuit consists of a combinational circuit to which storage elements are connected to form a feedback path



- The storage elements are devices capable of storing binary information
- The binary information stored in these elements at any given time defines the **state** of the sequential circuit at that time

Sequential Circuits (2/2)

- The outputs in a sequential circuit are a function not only of the inputs, but also of the present state of the storage elements
- The next state of the storage elements is also a function of external inputs and the present state
- Thus, ***a sequential circuit is specified by a time sequence of inputs, outputs, and internal states***

Types of Sequential Circuits

- There are two main types of sequential circuits, and their classification is a function of the timing of their signals
 - Synchronous Sequential Circuits
 - It is a system whose behavior can be defined from the knowledge of its signals at discrete instants of time
 - Asynchronous Sequential Circuits
 - The behavior of an asynchronous sequential circuit depends upon the input signals at any instant of time and the order in which the inputs change
 - Here we will discuss only the synchronous sequential circuits

Synchronous Sequential Circuits (1/3)

- A synchronous sequential circuit employs signals that affect the storage elements at only discrete instants of time
- Synchronization is achieved by a timing device called a **clock generator**, which provides a clock signal having the form of a periodic train of clock pulses



Timing diagram of a Clock Signal

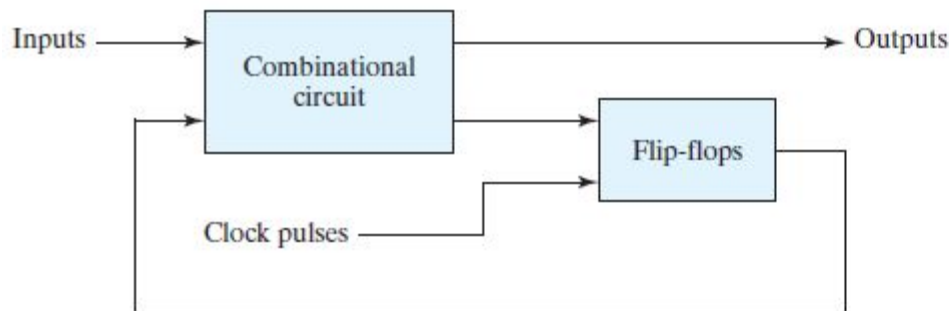
- The clock signal is commonly denoted by the identifiers **clock** and **clk**
- The clock pulses are distributed throughout the system in such a way that storage elements are affected only with the arrival of each pulse

Synchronous Sequential Circuits (2/3)

- In practice:
 - The clock pulses determine when computational activity will occur within the circuit
 - And other signals (external inputs and otherwise) determine what changes will take place affecting the storage elements and the outputs
- For example, a circuit that is to add and store two binary numbers:
 - Would compute their sum from the values of the numbers
 - And store the sum at the occurrence of a clock pulse

Synchronous Sequential Circuits (3/3)

- Synchronous sequential circuits that use clock pulses to control storage elements are called **clocked sequential circuits**
- They are called synchronous circuits because the activity within the circuit and the resulting updating of stored values is synchronized to the occurrence of clock pulses
- The storage elements (memory) used in clocked sequential circuits are called **flip flops**



Timing diagram of a Clock Signal

Storage Elements

- A storage element in a digital circuit can maintain a binary state indefinitely (as long as power is delivered to the circuit), until directed by an input signal to switch states
- There are two types of storage elements
 - Latches
 - Storage elements that operate with signal levels are referred to as latches
 - Flip Flops
 - Storage elements that are controlled by a clock transition are flip-flops

Latches

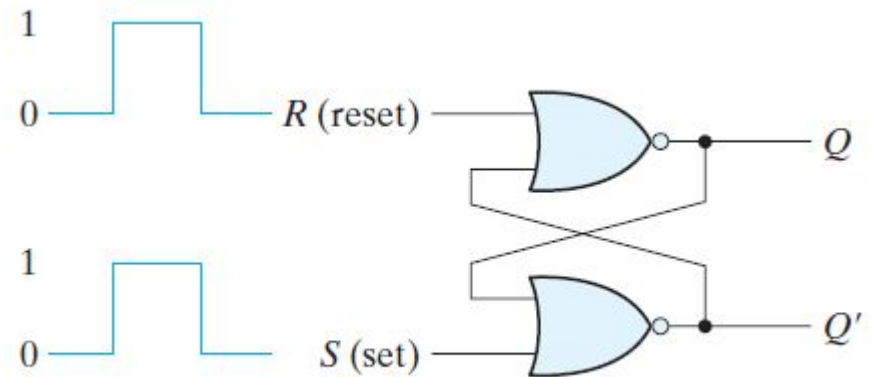
- Storage elements that operate with signal levels are referred to as latches
- Latches are said to be level sensitive devices
- Latches are used for the design of asynchronous sequential circuits
- The knowledge of latches is important as they are the basic circuits from which all flip-flops are constructed
- Here we will discuss two types of latches
 - SR Latch
 - D Latch

SR Latch

- The SR latch is a circuit with two cross-coupled NOR gates or two cross-coupled NAND gates
- Two inputs
 - S (stands for set)
 - R (stands for reset)
- Two outputs Q and Q' determine two possible states
 - If Q = 1 latch is in set state
 - If Q = 0 latch is in reset state

SR Latch with NOR Gate (1/2)

- Works on active high state
- It is mandatory to connect R with the gate producing Q and S with the gate producing Q'
- Case 1 ($S = 0$ $R = 0$)
 - No change in state
- Case 2 ($S = 0$ $R = 1$)
 - $Q = 0$ $Q' = 1$
 - Reset State
- Case 3 ($S = 1$ $R = 0$)
 - $Q = 1$ $Q' = 0$
 - Set State
- Case 4 ($S = 1$ $R = 1$)
 - $Q = 0$ $Q' = 0$
 - Practically Impossible
 - Indeterminate state



Logic diagram

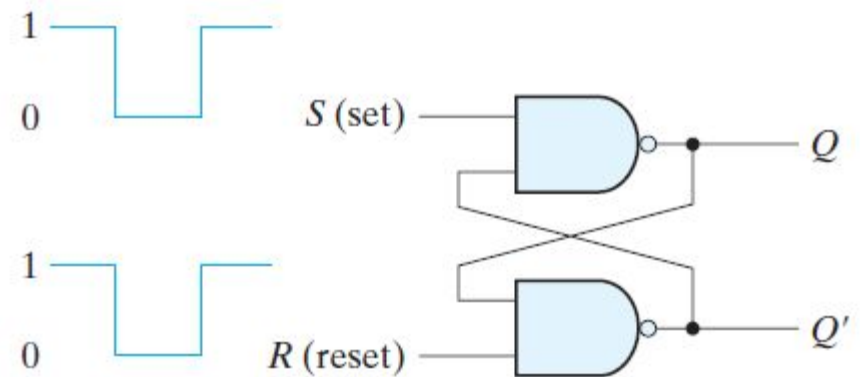
S	R	Q	State
0	0	Previous state	No Change
0	1	0	Reset
1	0	1	Set
1	1	?	Forbidden

SR Latch with NOR Gate (2/2)

- Under normal conditions, both inputs of the latch remain at 0 unless the state has to be changed
- The application of a momentary 1 to the S input causes the latch to go to the set state
- The S input must go back to 0 before any other changes take place, in order to avoid the occurrence of an undefined next state
- The application of a momentary 1 to the R input causes the latch to go to the Reset state

SR Latch with NAND Gate (1/2)

- Works on active low state
- It is mandatory to connect S with the gate producing Q and R with the gate producing Q'
- Case 1 ($S = 0$ $R = 0$)
 - $Q = 1$ $Q' = 1$
 - Practically Impossible
 - Indeterminate state
- Case 2 ($S = 0$ $R = 1$)
 - $Q = 1$ $Q' = 0$
 - Set State
- Case 3 ($S = 1$ $R = 0$)
 - $Q = 0$ $Q' = 1$
 - Reset State
- Case 4 ($S = 1$ $R = 1$)
 - No change in state



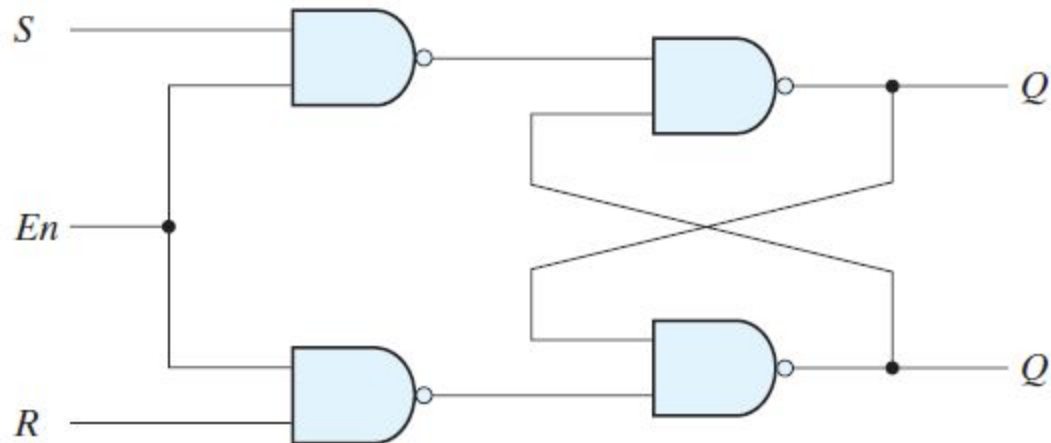
S	R	Q	State
0	0	?	Forbidden
0	1	1	Set
1	0	0	Reset
1	1	Previous State	No Change

SR Latch with NAND Gate (2/2)

- Also called S'R' Latch
- Under normal conditions, both inputs of the latch remain at 1 unless the state has to be changed
- The application of a momentary 0 to the S input causes the latch to go to the set state
- The S input must go back to 1 before any other changes take place, in order to avoid the occurrence of an undefined next state
- The application of a momentary 0 to the R input causes the latch to go to the Reset state

SR Latch with Enable Input (1/2)

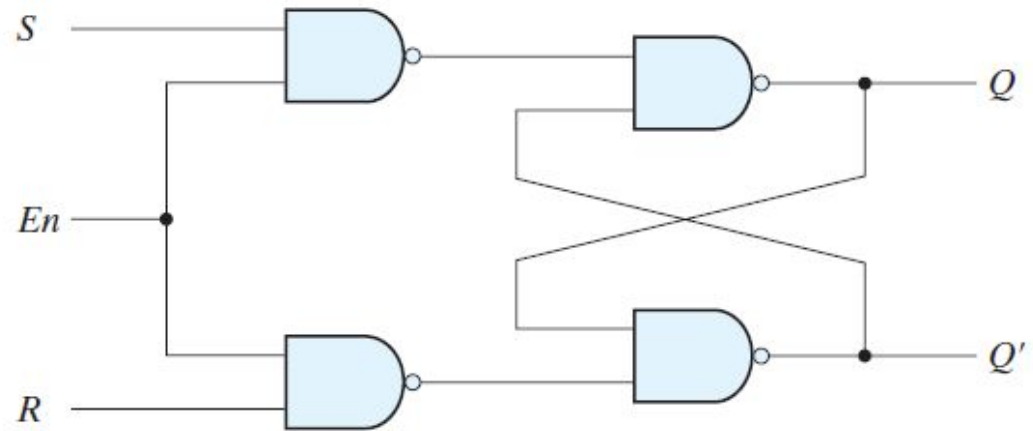
- The operation of the basic SR latch can be modified by providing an additional input signal that determines (controls) when the state of the latch can be changed by determining whether S and R can affect the circuit



- The outputs of the NAND gates stay at the logic-1 level as long as the enable signal remains at 0

SR Latch with Enable Input (2/2)

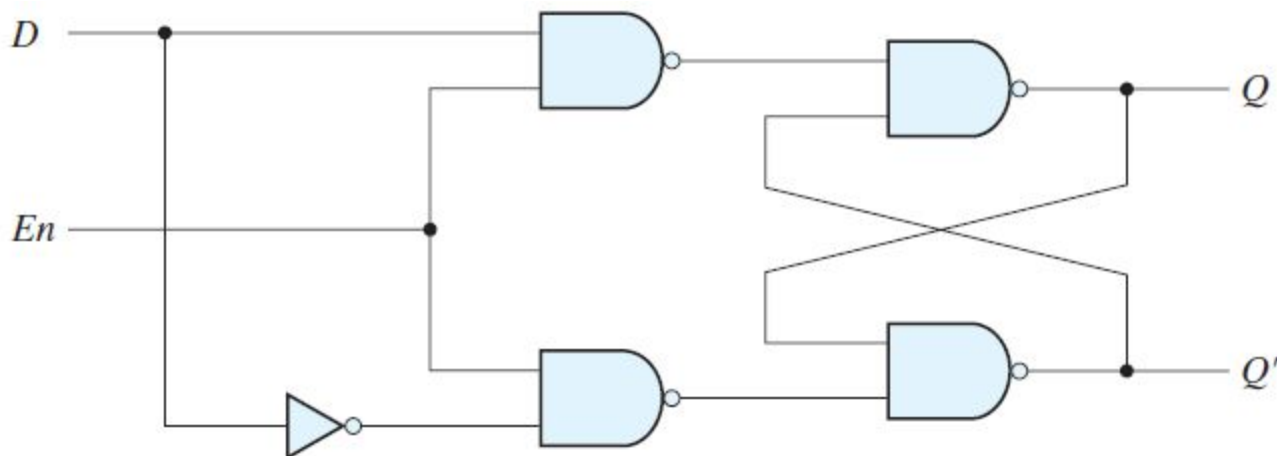
- $En = 0; S = R = X$
 - No change in state
- $En = 1; S = 0; R = 0$
 - No change in state
- $En = 1; S = 0; R = 1$
 - $Q = 0 \quad Q' = 1$
 - Reset State
- $En = 1; S = 1; R = 0$
 - $Q = 1 \quad Q' = 0$
 - Set State
- $En = 1; S = 1; R = 1$
 - $Q = 1 \quad Q' = 1$
 - Practically Impossible
 - Indeterminate state



En	S	R	Next State of Q
0	X	X	No Change
1	0	0	No Change
1	0	1	$Q = 0 \rightarrow$ Reset State
1	1	0	$Q = 1 \rightarrow$ Set State
1	1	1	Indeterminate state

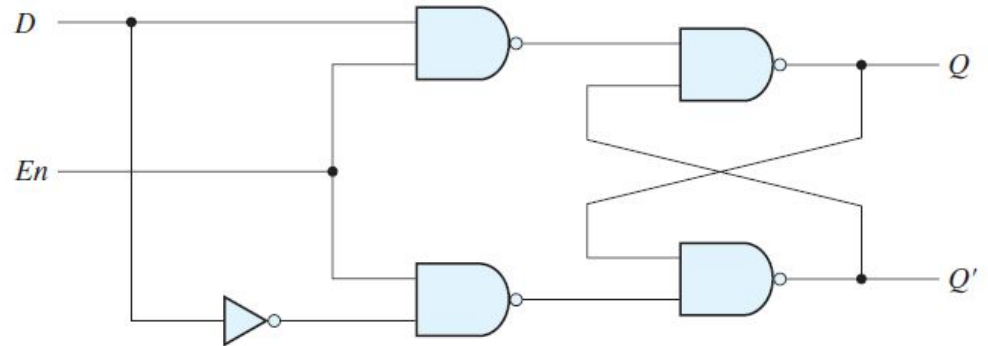
D Latch (1/2)

- One way to eliminate the undesirable condition of the indeterminate state in the SR latch is to ensure that inputs S and R are never equal to 1 at the same time
- This is done in the D latch (which is also called transparent latch)



D Latch (2/2)

- Case 1 ($En = 0$; $D = X$)
 - No change in state
- Case 2 ($En = 1$; $D = 0$)
 - $Q = 0$
 - Reset State
- Case 3 ($En = 1$; $D = 1$)
 - $Q = 1$
 - Set State



En	D	Next State of Q
0	X	No Change
1	0	$Q = 0 \rightarrow$ Reset State
1	1	$Q = 1 \rightarrow$ Set State

Graphical Symbols of Latches

- Inputs to the left
- Outputs to the right
- SR latch is constructed with NOR gates
- S'R' latch is constructed with NAND gate

